

Electronics MP1

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1 Circuit Design

The overall circuit takes 5V from USB, regulates it down to 3.3V, and uses the 3.3V supply to power a hysteretic oscillator. A voltage divider and unity-gain follower provide a stable 1.65V reference for the oscillator. The oscillator output then drives an LED, producing a flash period of approximately 1 second. This is done entirely with analog circuitry, without any digital logic or external clock. The complete circuit schematic is shown below.

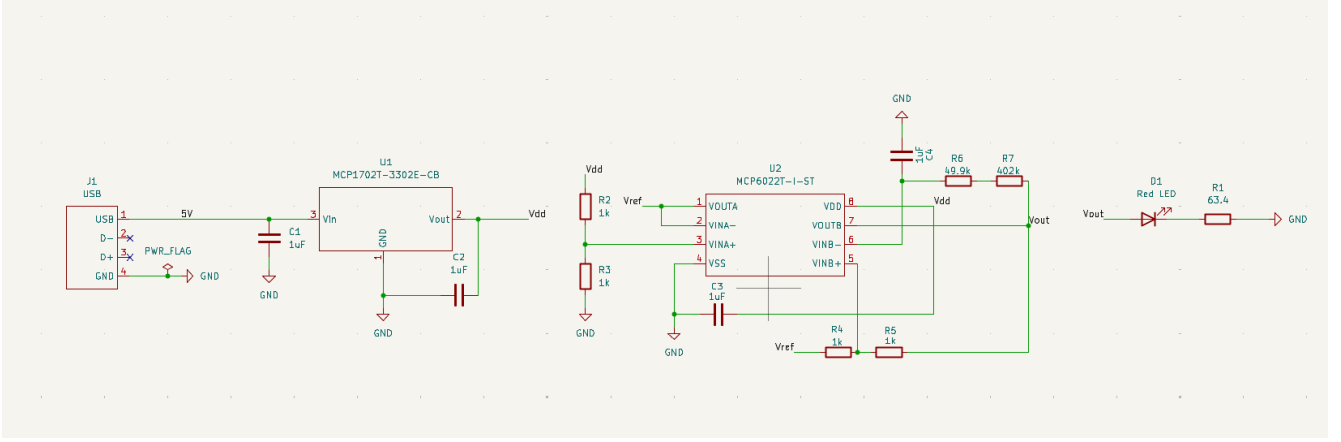


Figure 1: Complete circuit schematic for the USB-powered LED flasher.

2 Parts Selection

I first went through the provided parts list and datasheets to choose the main components needed for the circuit. I selected the provided USB connector, the MCP1702T-3302E/CB 3.3V LDO, the MCP6022T-I/ST dual op-amp, the LTST-C171KRKT red LED, and appropriate resistors and capacitors. The dual op-amp was useful because I could use one op-amp for buffering the voltage reference and the other for the hysteretic oscillator.

3 USB Input and LDO

The circuit is powered from the 5V USB VBUS and GND pins, while the two USB data pins are left unconnected. VBUS is connected to the input of the MCP1702T-3302E/CB 3.3V LDO, which I wired according to its datasheet. Following the datasheet recommendations, I also included two 1 μF bypass capacitors, one at V_{IN} and one between V_{DD} and GND.

4 1.65 V Reference

Since the oscillator operates between 0 V and 3.3 V, I chose the midpoint of the supply as the reference voltage:

$$V_{\text{ref}} = \frac{3.3}{2} = 1.65 \text{ V}.$$

I generated this reference using two equal 1 k Ω resistors as a voltage divider:

$$V_{\text{ref}} = 3.3 \left(\frac{1 \text{ k}\Omega}{1 \text{ k}\Omega + 1 \text{ k}\Omega} \right) = 1.65 \text{ V}.$$

I then used one half of the MCP6022T-I/ST as a unity-gain voltage follower. The follower does not change the reference voltage, but buffers the voltage divider so that loading from the oscillator does not significantly change V_{ref} .

5 Hysteretic Oscillator

The second half of the MCP6022T-I/ST is used to create an inverting hysteretic oscillator. The positive-feedback resistor network sets the Schmitt-trigger switching thresholds, while the timing resistor and capacitor determine how quickly the circuit moves between those thresholds.

I chose the Schmitt-trigger feedback resistances to have a 1:1 ratio, giving

$$\alpha = 0.5.$$

With $V_{DD} = 3.3 \text{ V}$ and $V_{\text{ref}} = 1.65 \text{ V}$, this produces symmetric switching thresholds of

$$V_{\text{up}} = \alpha V_{\text{ref}} = (0.5)(1.65) = 0.825 \text{ V},$$

and

$$V_{\text{dn}} = V_{DD} - \alpha(V_{DD} - V_{\text{ref}}) = 3.3 - 0.5(3.3 - 1.65) = 2.475 \text{ V}.$$

These thresholds are symmetric around 1.65 V, so the ideal charging and discharging times are equal, resulting in approximately a 50% duty cycle.

When the oscillator output is HIGH, the timing capacitor charges through the timing resistor until its voltage reaches 2.475 V. The Schmitt trigger then switches the output LOW. The capacitor then discharges until it reaches 0.825 V, at which point the output switches HIGH again. This process repeats continuously to create the oscillation.

6 Oscillator Period

For the discharging portion of the cycle,

$$T_l = RC \ln \left(\frac{V_{\text{dn}}}{V_{\text{up}}} \right).$$

Using the selected thresholds,

$$T_l = RC \ln \left(\frac{2.475}{0.825} \right) = RC \ln(3) = 1.0986RC.$$

Because the thresholds are symmetric, the charging time is the same:

$$T_h = 1.0986RC.$$

The total period is therefore

$$T = T_l + T_h$$

and

$$T = 2(1.0986RC) = 2.197RC.$$

For a target period of 1 second, the required RC time constant is

$$RC = \frac{1}{2.197} \approx 0.455 \text{ s.}$$

I selected a timing capacitor of

$$C = 1 \text{ } \mu\text{F}$$

and a timing resistance of

$$R = 451.9 \text{ k}\Omega.$$

This gives

$$RC = (451.9 \text{ k}\Omega)(1 \text{ } \mu\text{F}) = 0.4519 \text{ s.}$$

The resulting nominal period is

$$T = 2.197(0.4519) \approx 0.993 \text{ s,}$$

which is very close to the desired 1 second period.

7 Simulation and Component Tolerances

Since the calculated nominal period was very close to the desired 1 second period, I was satisfied with the selected RC values. I then modeled the oscillator and unity-gain follower in LTSpice to determine how component variation could affect the period. I used the specified component tolerances of $\pm 1\%$ for the resistors and $\pm 10\%$ for the $1\ \mu\text{F}$ capacitor. These tolerances were modeled using Gaussian variations in LTSpice, and I ran 250 simulations with randomly varied component values.

The LTSpice setup used for this tolerance analysis is shown in Figure 2.

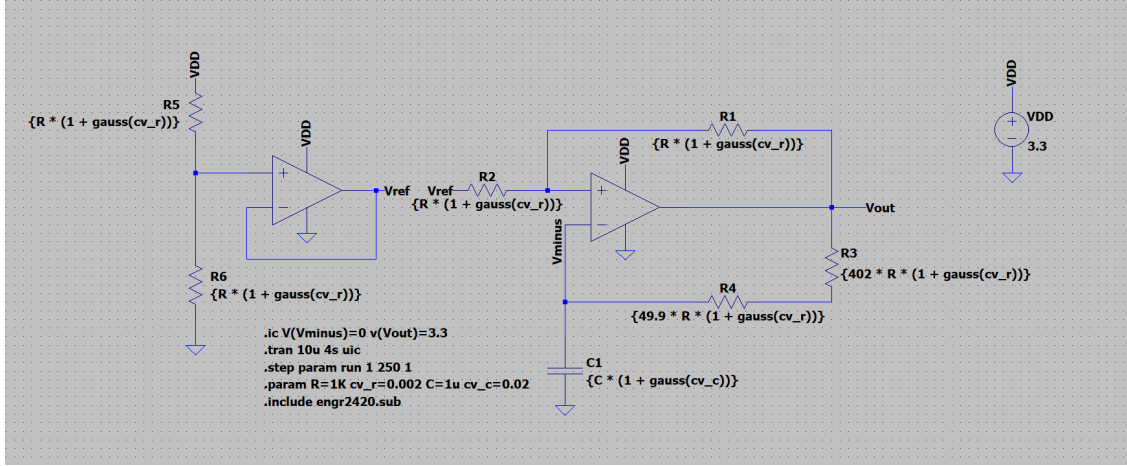


Figure 2: LTSpice setup used for the component tolerance simulations.

I then ran the LTSpice simulation and probed V_{out} , producing the oscillating output waveform shown in Figure 3.

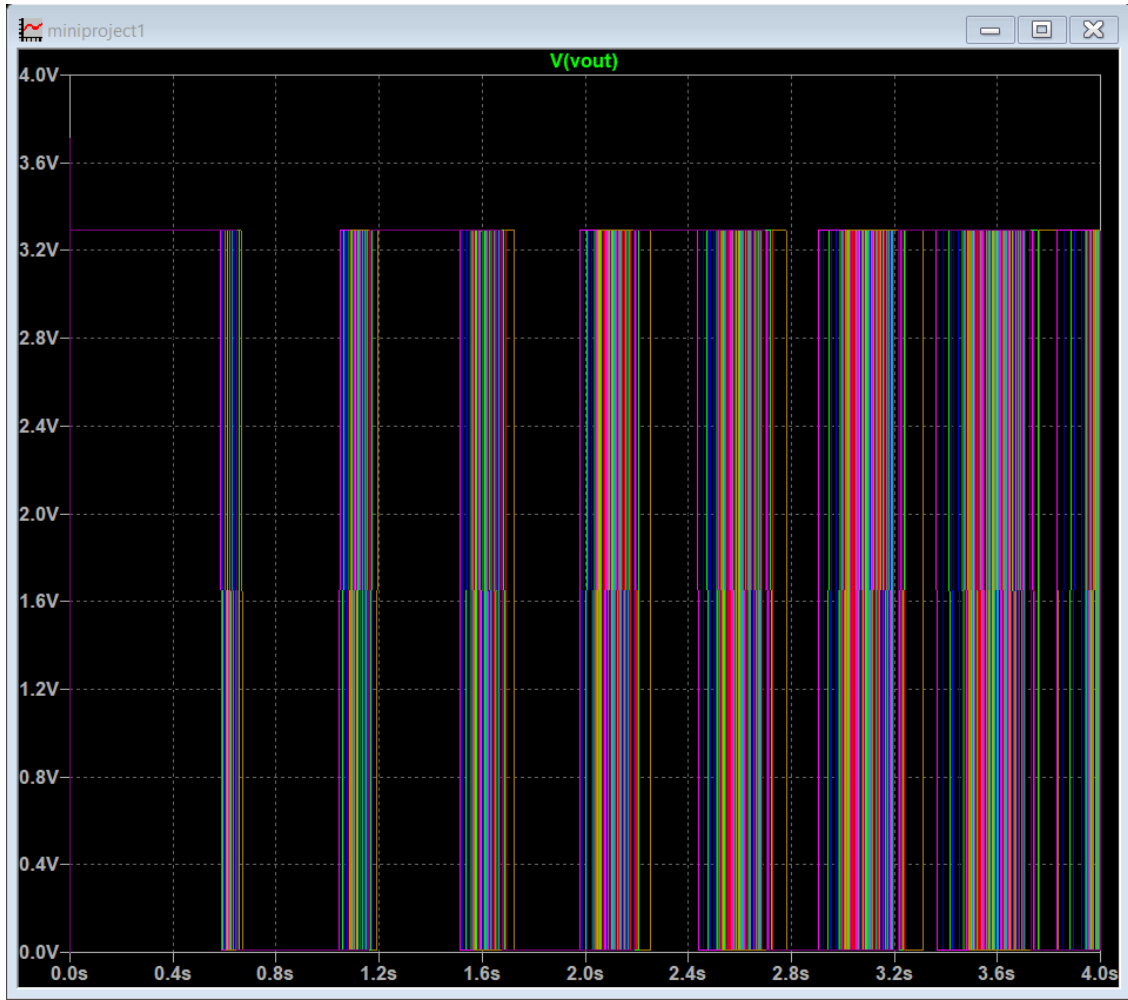


Figure 3: Example oscillator output waveform from LTSpice.

I then processed the output waveforms from all 250 simulations in MATLAB. For each simulation run, the script identified the rising edges of V_{out} and calculated the time between consecutive rising edges to determine each oscillation period. These period measurements were then averaged within each run, giving one average period for each of the 250 simulations. The distribution of these average periods is shown in Figure 4.

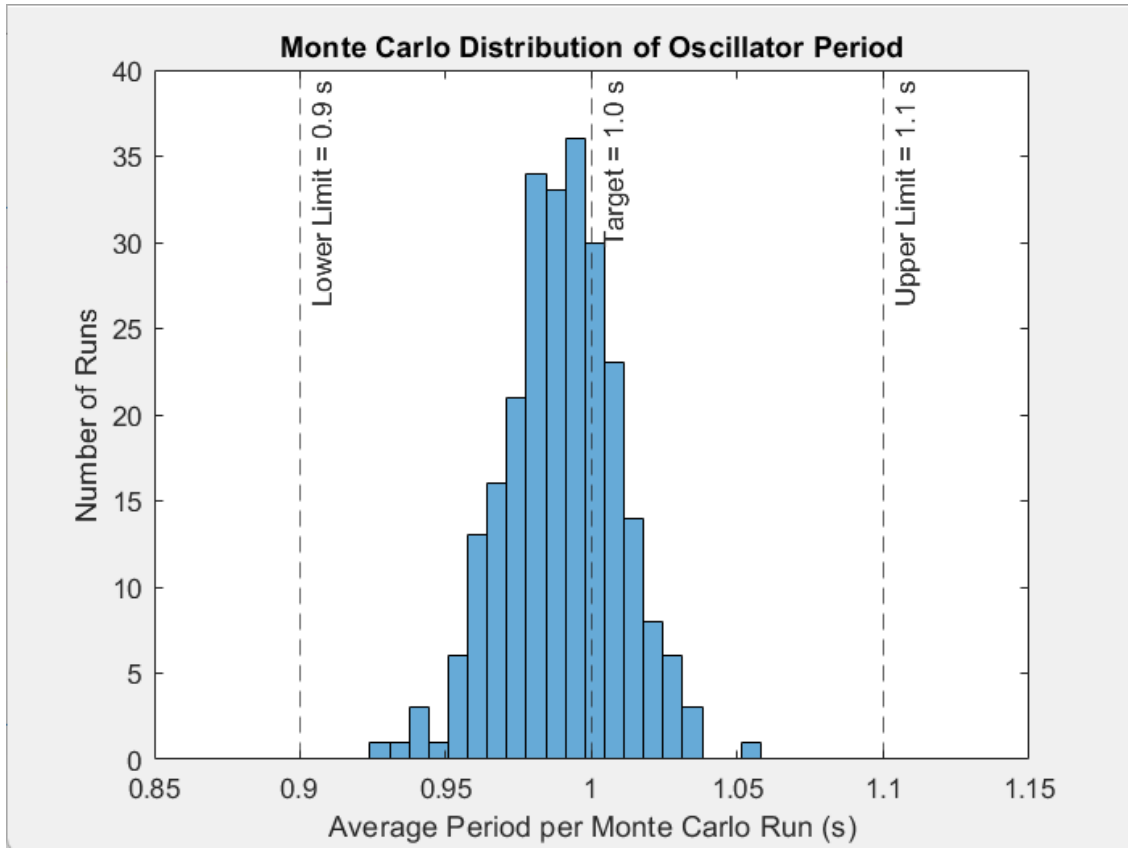


Figure 4: Distribution of the average oscillator period across the 250 Monte Carlo simulations.

I then compared the average period from each simulation against the required $\pm 10\%$ tolerance around the desired 1 second period. This corresponds to an acceptable period range of 0.9–1.1 seconds. Figure 5 shows the average period from each of the 250 simulations along with the target period and tolerance limits.

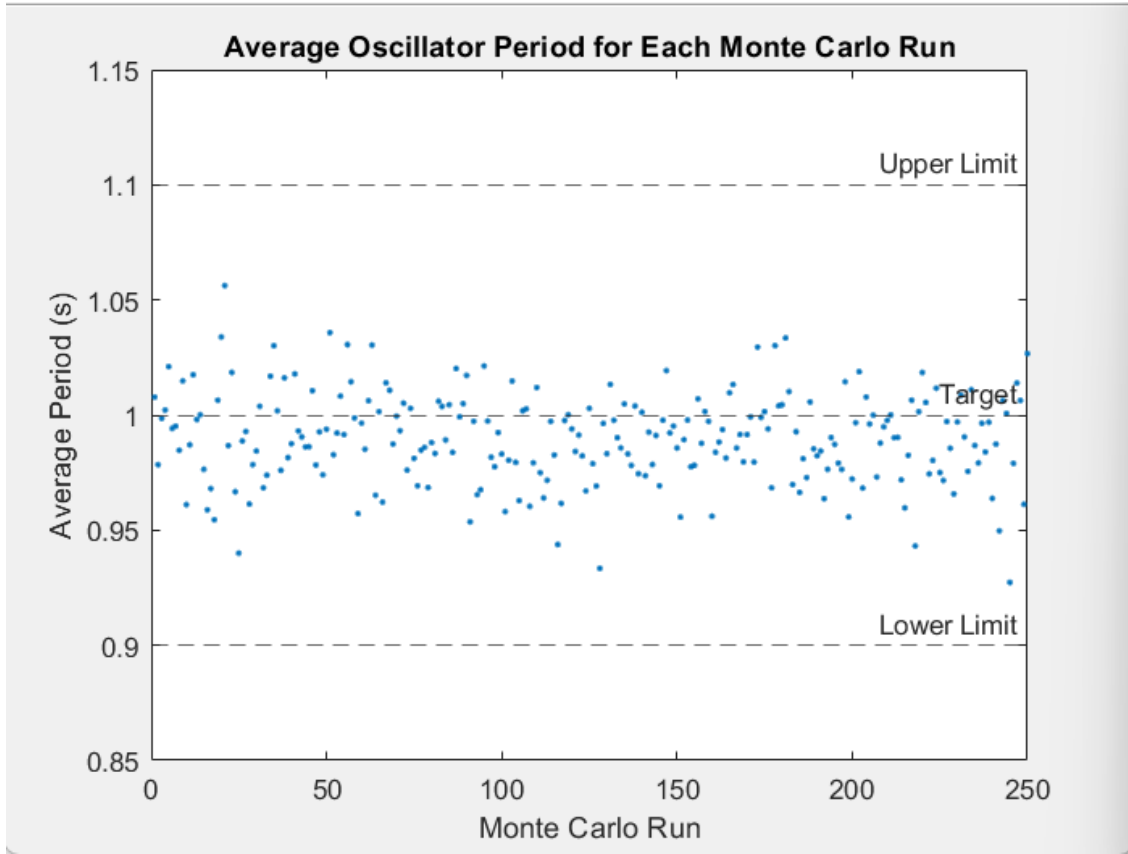


Figure 5: Average oscillator period for each of the 250 Monte Carlo simulations, with the 1 second target and $\pm 10\%$ limits shown.

All 250 simulated average periods remained within the required 0.9–1.1 second range. Because the oscillator and voltage-reference circuitry determine the timing of the LED flasher, I assumed that if this subsystem met the period requirement under component variation, the complete circuit would also meet the required period specification.

8 LED Output

The oscillator output drives an LTST-C171KRKT red LED through a current-limiting resistor. I targeted an LED current of approximately 20 mA. From the LED’s datasheet provided I–V curve, I estimated a forward voltage of approximately

$$V_F \approx 2.05 \text{ V}$$

at 20 mA.

The voltage across the current-limiting resistor is therefore approximately

$$V_R = 3.3 - 2.05 = 1.25 \text{ V}.$$

The required resistance is

$$R = \frac{V_R}{I_{\text{LED}}} = \frac{1.25}{0.020} = 62.5 \, \Omega.$$

I selected the closest available resistor value,

$$R_{\text{LED}} = 63.4 \, \Omega.$$

This gives an estimated LED current of

$$I_{\text{LED}} = \frac{3.3 - 2.05}{63.4} \approx 19.7 \, \text{mA}.$$

Therefore, the $63.4 \, \Omega$ current-limiting resistor keeps the estimated LED current close to the desired 20 mA when the oscillator output is HIGH.